# Chunhui Zhang, CS Ph.D. candidate at Dartmouth

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## Research Focus

**RL Multimodal LLMs** on Language, vision, and audio

## Education

**Dartmouth College** Hanover, NH, USA
Ph.D. candidate in Computer Science Aug. 2023–Jun. 2026 (expected)
Advisor: Professor Soroush Vosoughi

**Brandeis University** Waltham, MA, USA
M.S. in Computer Science Sep. 2021–Jun. 2023
GSAS Research Fellowship

**Northeastern University** China
B.S. in Computer Science Sep. 2017–Jun. 2021
Outstanding Honor Thesis Award

## Experience

**Amazon** Seattle, WA, USA
Applied Scientist Intern Sep. 2025 – Dec. 2025
*RL Post-Training for Multimodal Computer-Use Agents (CUA) in Rufus (reorganized into Amazon AGI Nova prior to internship end)*
Post-trained multimodal LLMs (7–32B) for low-latency GUI agents towards **adaptive reasoning** with reduced token costs.
Developed verl and vLLM to enable **on-policy RL** and **SFT** on complex trajectories, bridging OOD gaps and stabilizing visual grounding.

**Google DeepMind** Mountain View, CA, USA
Research Intern Jun. 2025 – Sep. 2025
*Gemma 3n Post-Training for Real-Time Streaming (Frontier AI & GenAI)*
Created a synthetic audio–text framework for real-time ASR/AST (speech/voice tasks); eliminated human labeling dependence while keeping WER/BLEU within **1%** of baselines.
Built the **full-stack RL+SFT infrastructure** for Gemma 3n, enabling PyTorch/SGLang support for the open-source community.
Engineered a high-throughput training pipeline for Gemma 3n, accelerating RL (GRPO, DAPO) by 5×.
Contributed to the **Gemma 3n technical report** and open-source development.

**Honda Research Institute USA** San Jose, CA, USA
Research Intern Jun. 2024–Sep. 2024
*Multimodal LLM Post-Training (8B–70B) for Social Reasoning*
Published a *spotlight* paper at ICML 2025 (Top 2.59%). Filed a US patent. Developed long-context multimodal LLMs and accelerated inference-time scaling using vLLM framework. Enabled long-context modeling of human behavior and autonomous driving in multimodal environments; scalable to LLMs up to 70B/405B parameters.

## Selected Publications

**AdaptGUI: Adaptive Reasoning and Reward Shaping for Efficient GUI Agents**
**Chunhui Zhang**, etc
*Under reivew*
Paper | Code
Post-trained multimodal LLMs (7–32B) for low-latency GUI agents towards **adaptive reasoning** with reduced token costs.

**Overcoming Multi-step Complexity in Theory-of-Mind Reasoning: A Scalable Bayesian Planner**
**Chunhui Zhang**, Zhongyu Ouyang, Sean Dae Houlihan, Kwonjoon Lee, Nakul Agarwal, Soroush Vosoughi, Shao-Yuan Lo
*ICML 2025* ***(Spotlight, Top 2.59%)***
Paper | Code
First scalable solution for multi-step Theory-of-Mind reasoning. Uses Bayesian inverse planning for global planning, then lets LLMs focus on local reasoning. Works on 70B+ models where others fail.

**Growing Through Experience: Scaling Episodic Grounding in Language Models**
**Chunhui Zhang**, Elsie Wang, Zhongyu Ouyang, Xiangchi Yuan, Soroush Vosoughi
*ACL 2025 Main Conference* ***(Oral Presentation, Top 3.24%)***
Paper | Code
Post-trained reasoning LLMs (across 3B, 8B, 70B) on MCTS-sampled data from physical simulators.

**Pretrained Image-Text Models are Secretly Video Captioners**
**Chunhui Zhang***, Yiren Jian*, Zhongyu Ouyang, Soroush Vosoughi
*NAACL 2025 Main Conference* ***(Oral Presentation, Top 2.88%)***
Paper | Code

RL post-training recipe that achieved **Top-2** on PapersWithCode video captioning leaderboard, outperforming industry MLLM captioners.

**Knowing More, Acting Better: Hierarchical Representation for Embodied Decision-Making**
**Chunhui Zhang**, Zhongyu Ouyang, Zheyuan Liu, Soroush Vosoughi
*EMNLP 2025 Findings*
Paper
Refines vision-language-action LLM representations to enable more effective PPO-based RL training in embodied AI.

**Working Memory Identifies Reasoning Limits in Language Models**
**Chunhui Zhang**, Yiren Jian, Zhongyu Ouyang, Soroush Vosoughi
*EMNLP 2024 Main Conference* | Paper | Code
Introduced working memory as a diagnostic tool for LLM reasoning limits. This work inspired a follow-up NAACL paper on long-context multimodal understanding.

**Temporal Working Memory: Query-Guided Segment Refinement for Enhanced Multimodal Understanding**
Xingjian Diao*, **Chunhui Zhang***, Weiyi Wu, Zhongyu Ouyang, Peijun Qing, Ming Cheng, Soroush Vosoughi, Jiang Gui
*NAACL 2025 Findings* | Paper | Code
Inspired by working memory in my EMNLP 2024 paper, this work is a follow-up study on long-context video-language understanding.

## Under Review

**Model Priors Shape Experience: RL for Complex Audio Long-form Reasoners**
**Chunhui Zhang**
Code
*First to implement Qwen2.5-Omni and vLLM for faster RL reasoning across audio and other unified modalities.*

## Selected Collaborations

**Superficial Self-Improved Reasoners Benefit from Model Merging**
Xiangchi Yuan, **Chunhui Zhang**, Zheyuan Liu, Dachuan Shi, Soroush Vosoughi, Wenke Lee
*EMNLP 2025* | Paper | Code

**SoundMind: RL-Incentivized Logic Reasoning for Audio-Language Models**
Xingjian Diao, **Chunhui Zhang**, Keyi Kong, Weiyi Wu, Chiyu Ma, Zhongyu Ouyang, Peijun Qing, Soroush Vosoughi, Jiang Gui
*EMNLP 2025* ***(Oral Presentation)***

**Is It Navajo? Accurate Language Detection in Endangered Athabaskan Languages**
Ivory Yang, Weicheng Ma, **Chunhui Zhang**, Soroush Vosoughi
*NAACL 2025* ***(Oral Presentation, Top 2.88%)*** | Paper | Code

**Learning Musical Representations for Music Performance Question Answering**
Xingjian Diao, **Chunhui Zhang**, Tingxuan Wu, Ming Cheng, Zhongyu Ouyang, Weiyi Wu, Jiang Gui
*EMNLP findings 2024*

**Expedited Training of Visual Conditioned Language Generation via Redundancy Reduction**
Yiren Jian, Tingkai Liu, Yunzhe Tao, **Chunhui Zhang**, Soroush Vosoughi, Hongxia Yang
*ACL 2024* ***(Oral Presentation, Top 3.10%)***

**Aligning Relational Learning with Lipschitz Fairness**
Yaning Jia*, **Chunhui Zhang***, Soroush Vosoughi
*ICLR 2024 – Note: Co-first author Jia was a master student I mentored.*

**Mitigating Emergent Robustness Degradation on Graphs while Scaling-up**
Xiangchi Yuan*, **Chunhui Zhang***, Yijun Tian, Yanfang Ye, et al.
*ICLR 2024 – Note: Co-first author Yuan was a master student I mentored.*

**When Sparsity Meets Contrastive Models: Less Data Can Bring Better Class-Balanced Representations**
**Chunhui Zhang**, Chao Huang, Yijun Tian, Qianlong Wen, et al.
*ICML 2023* – **AAAI-DCAA 2023 Best Paper Runner-up Award**

**Chasing All-Round Graph Representation Robustness: Model, Training, and Optimization**
**Chunhui Zhang**, Yijun Tian, Mingxuan Ju, Zheyuan Liu, et al.
*ICLR 2023*

**Look Twice as Much as You Say: Scene Graph Contrastive Learning for Self-Supervised Image Caption Generation**
**Chunhui Zhang**, Chao Huang, Youhuan Li, Xiangliang Zhang, Yanfang Ye, et al.
*CIKM, 2022.*

***Full publication list*** *includeing 30+ papers at ICML, NeurIPS, ICLR, ACL, NAACL, EMNLP, WWW (TheWebConf), KDD, CIKM, ICDM, ICWSM, etc. available at chunhuizng.github.io*

## Honors and Awards

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|---|---|
| ICML Spotlight Recognition (Top 2.59%) | 2025 |
| ACL Oral Presentation Award (Top 3.24%) | 2024, 2025 |
| NAACL Oral Presentation Award (Top 2.88%) | 2025 |
| AAAI-DCAA Best Paper Runner-up Award | 2023 |
| Graduate School of Arts and Sciences Fellowship | 2021–2023 |

## Service

**Program Committee/Reviewer:** ICML, ICLR, NeurIPS, ACL, NAACL, EMNLP, UAI, AISTATS, AAAI (2023–2025)
**Journal Reviewer:** TMLR, IEEE TKDE, IEEE TNSE, ACM TIST, Neurocomputing